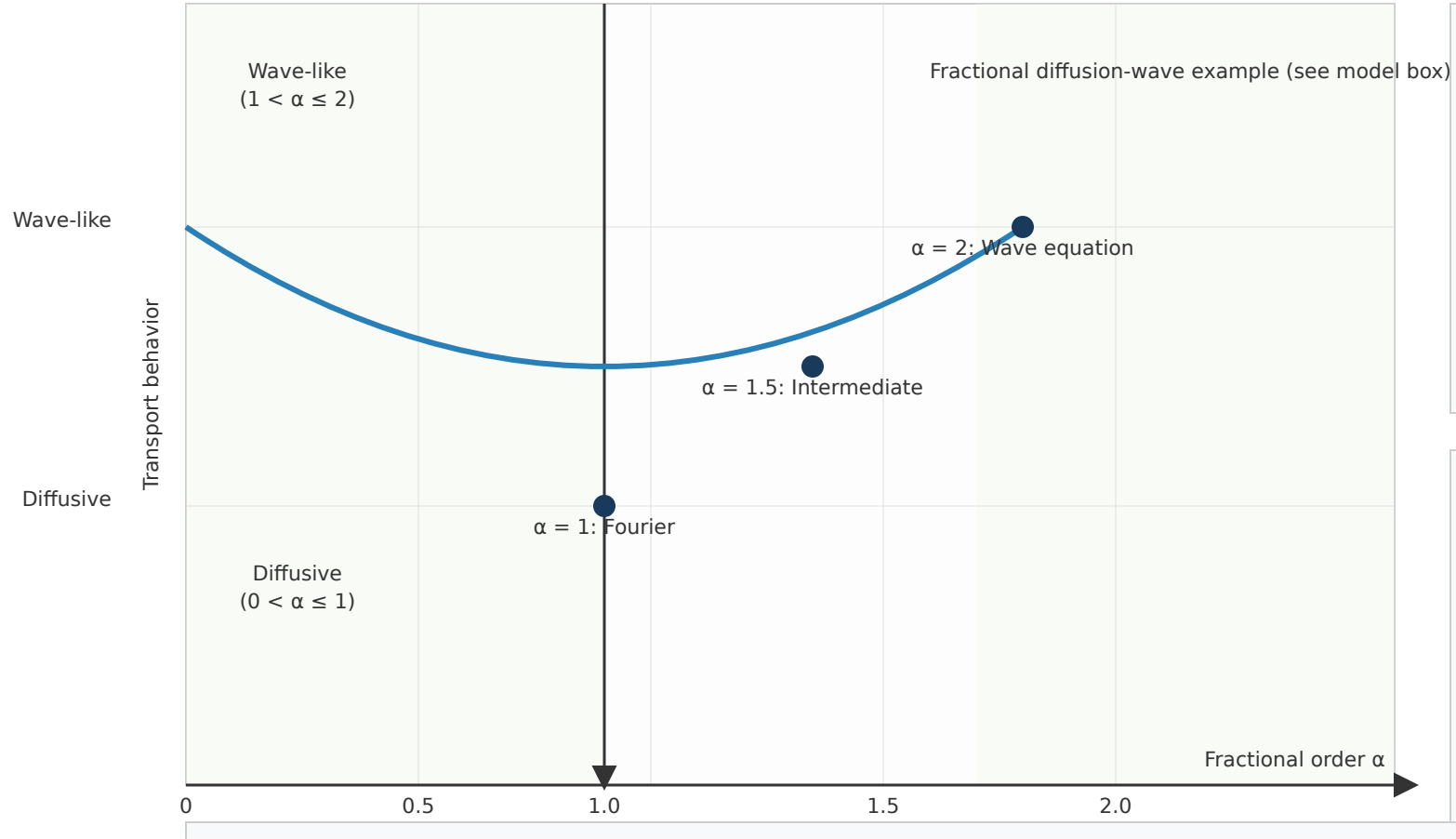
Fractional derivative order and transport behavior

Interpolation between diffusive and wave-like heat transport (Sections 6, 11)



Fractional models

α = 1: Classical Fourier diffusion
{}^C DT = a∇T

1 < α ≤ 2: Wave-like heat propagation
{}^C DT = a∇T

α = 2: Classical wave equation
∂T/∂t = a∇T

0 < α < 1: Sub-diffusive

Key equations

Eq. (6.3) Fractional heat equation
Eq. (11.1) Fractional thermoelasticity
Eq. (11.2) Fractional Cattaneo equation
Eq. (6.1) Caputo derivative
Eq. (6.2) Riemann-Liouville derivative

Physical interpretation

Fractional derivatives introduce memory: the response at time t depends on the history of the stimulus.
The order α controls the interpolation between diffusive (α = 1) and wave-like (α = 2) behavior.
Propagation/support depends on the fractional model; no family-wide finite-speed claim.
Caputo derivative preferred for physical applications (classical initial conditions).
Well-posedness requires model- and space-specific proof.